```
[mysqld]
NodeId=8
hostname=172.22.44.100

[mysqld]
NodeId=9
hostname=172.22.33.100

[mysqld]
NodeId=10
hostname=172.22.22.100
```

Note: It is recommended that the MySQL root password is set before proceeding.

MySQL cluster set-up

On management nodes: To start the cluster, use the following command:

```
ndb_mgmd -f /var/lib/mysql-cluster/config.ini
```

It should show the following output:

```
MySQL Cluster Management Server mysql-5.6.27 ndb-7.3.11
2016-01-13 11:21:54 [MgmtSrvr] INFO     -- The default config
directory '/usr/mysql-cluster' does not exist. Trying to
create it...
2016-01-13 11:21:54 [MgmtSrvr] INFO     -- Sucessfully
created config directory
```

On data nodes: To start the NDB cluster engine, use the following command:

```
ndbd
```

It should show the output given below:

```
2016-01-13 11:22:40 [ndbd] INFO
-- Angel connected to '172.22.100.10:1186'
2016-01-13 11:22:40 [ndbd] INFO     -- Angel allocated
nodeid: 3
```

On MySQL nodes: To start the MySQL service, use the command shown below:

```
service mysql start
```

To check the cluster status from the management node, use the following command:

```
ndb_mgm
show
```

If the cluster is healthy, it must display the following output:

```
mgmtsrv02 root [mysql-cluster] > ndb_mgm
-- NDB Cluster -- Management Client --
ndb_mgm> show
Connected to Management Server at: 172.22.100.10:1186
Cluster Configuration
---------------------
[ndbd(NDB)]     4 node(s)
id=3 @172.22.11.100 (mysql-5.6.27 ndb-7.3.11,Nodegroup: 0, *)
id=4  @172.22.44.100  (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0)
id=5  @172.22.33.100  (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0)
id=6  @172.22.22.100  (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0)

[ndb_mgmd(MGM)] 2 node(s)
id=1    @172.22.100.10  (mysql-5.6.27 ndb-7.3.11)
id=2    @172.22.96.10  (mysql-5.6.27 ndb-7.3.11)

[mysqld(API)]   4 node(s)
id=7    @172.22.11.100  (mysql-5.6.27 ndb-7.3.11)
id=8    @172.22.44.100  (mysql-5.6.27 ndb-7.3.11)
id=9    @172.22.33.100  (mysql-5.6.27 ndb-7.3.11)
id=10   @172.22.22.100  (mysql-5.6.27 ndb-7.3.11)
```

Otherwise, in case if any node goes down, it would show:

```
[ndbd(NDB)]     4 node(s)
id=3 @172.22.11.100 (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0, *)
id=4  @172.22.44.100  (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0)
id=5 (not connected, accepting connect from 172.22.33.100)
id=6  @172.22.22.100  (mysql-5.6.27 ndb-7.3.11, Nodegroup: 0)
```

Any DBMS uses storage engines or database engines to write, read, update or delete data from a database. InnoDB is the default storage engine used by MySQL since version 5.5, such that whenever a table is created without the ENGINE clause, it creates an InnoDB table, by default. With InnoDB, data is read and written from the hard disk, where the MySQL server runs, and hence it necessitates configuring the disks in RAID, to achieve data redundancy.

On the other hand, the MySQL cluster uses the NDBCluster engine, which uses network connectivity in order to access data spread across different data nodes (not on MySQL servers like InnoDB). Hence, while creating tables, one must explicitly mention the NDBCluster storage engine in order to instruct MySQL servers that data has to be stored on the data nodes. END

By: Mandar Shinde

The author works in the IT division of one of the largest commercial automotive organisations in India. His technical interests include Linux, networking, backups and virtualisation.